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1. **What are some of the applications of DE in the real world?**

   There are many applications of Differential Equations in the real world, especially in technologies. As our prof had mentioned, DE can be used to solve all kinds of problems from calculating how a cup of coffee cools down to determining the time of death of a corpse. But aside from these, DE can also be used in things that we use everyday especially in our current devices and technologies, like calculating how fast a battery loses its charge, how heat affects a computer processor or even how electricity flows in a circuit. Differential Equations can also be used to predict what will happen in the future based on how something changes over time. For me, it's interesting how something that we usually just solve on paper can actually be applied to real-life situations, even in the technologies that we use everyday without really thinking about how they work.

2. **How can DE model the spread of disease in a pandemic?**

   Differential Equations can be used to model how a disease spreads in a population over time. For example, it can help us estimate how many people might become infected, recover or are still at risk of getting the disease. One of the DE models that can be used to model the spread of a disease is the **SIR model**. This model divides people into three groups, which are:

   - Susceptible - people who can get infected
   - Infected - people who currently have the disease
   - Recovered - people who have recovered

   The SIR model uses the equation $\frac{dI}{dt} = \beta\frac{SI}{N} - \gamma I$ to describe how the number of infected people changes over time, where (S) is the number of susceptible people, (I) is the number of infected people, (N) is the total population, (β) represents the transmission rate and (γ) represents the recovery rate. Through this, researchers can predict how fast a disease might spread or if the number of infected people will decrease. This can also help the government and health workers prepare hospitals and decide on possible ways to control the spread of the disease.

3. **How can DE model electronic circuits with possible components?**

The behavior of electronic components such as capacitors and inductors can be described using Differential Equations because their voltage and current can change over time. For example, in a resistor and capacitor (RC) circuit, the equation $RC\frac{dVc}{dt} + Vc = Vs$ can show how the capacitor's voltage changes while charging. Meanwhile, in a resistor and inductor (RL) circuit, the equation $L\frac{di}{dt} + Ri = Vs$ can describe how the current changes over time in this particular circuit. The inductor is related to the formula involving inductance (L), while the capacitor uses capacitance (C). These equations are useful in designing power supplies, filters and other electronic devices because they help us understand how the components respond to changes in voltage and current.

4. **Give a sample DE model and explain how it works and its uses.**

One example of a DE model used in robotics is the **DC motor model**, which describes how a DC motor converts electrical energy into mechanical movement. This model consists of two parts, which are the electrical model and the mechanical model.

The **electrical model** uses Kirchhoff's Voltage Law (KVL) to describe how the current changes inside the motor. Its equation is $V = Ri + L\frac{di}{dt} + K_b\omega$ where (V) is the applied voltage, (R) is the resistance, (i) is the current, (L) is the inductance, and $(K_b\omega)$ represents the back electromotive force or back EMF produced when the motor rotates. It explains how the current behaves or changes when voltage is applied to the motor.

Meanwhile, the **mechanical model** uses Newton's second law for rotational motion to describe how the motor rotates. Its equation is $J\frac{d\omega}{dt} = K_t i - b\omega$ where (J) is the rotational inertia, (ω) is the angular speed, (K_t) is the motor torque constant, (i) is the current and (b) represents the viscous friction coefficient. This equation explains how the motor's speed changes depending on the torque produced and the friction opposing the rotation.

These two equations work together because the electrical model describes the current that produces the motor's torque, while the mechanical model describes how that torque affects the motor's movement. This model is useful in robotics because motors are

used to move robot wheels, arms, and other moving parts. By using Differential Equations, engineers can predict how a motor responds to changes in voltage and control its speed and movement more accurately.

Reference:

MathWorks. (n.d.). DC Motor. Retrieved from https://www.mathworks.com/help/simscape-electrical/ref/dcmotor.html